

curriculum vitae

Materials Engineer with extensive experience in multi-scale materials modeling and experimental validation. Specializes in bridging microstructure to functional behavior, with a focus on nuclear reactor materials under extreme conditions. Combines deep computational mechanics expertise with hands-on experimental work. Effective communicator who bridges detailed scientific analysis with real-world application needs in multidisciplinary teams.

Personalia

Surname: Ruybalid
Given names: Andre, Paul
Degrees: Ph.D. | MSc. | BSc.
City: Los Alamos, New Mexico, U.S.A.
Phone number: +1 719 210 1746 (personal mobile)
E-mail: andreruybalid@gmail.com
Date of birth: 26 October 1987
Place of Birth: Farmington, New Mexico, U.S.A.
Nationality: American/Dutch

Professional experience

2023 - Present

R&D Engineer, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Developed multiple data-driven surrogate models and custom Python-based tools to bridge polycrystalline mechanics of nuclear reactor materials to engineering-scale predictions, enabling rapid materials selection and evaluation for reactor systems.
- ❖ Coordinated with mechanistic modelers as part of a larger multi-scale modeling workflow, ensuring consistent material predictions across length scales, directly applicable to materials selection for advanced reactor systems.
- ❖ Built and maintained material models for multi-physics, nuclear fuel performance analysis in MOOSE/BISON, including code contributions, verification, validation, and large-scale uncertainty quantification (UQ).
- ❖ Collaborated with external industrial partners and national lab colleagues to deliver pragmatic modeling solutions; led technical milestone reports, peer-reviewed papers, and presented the surrogate models at scientific conferences.
- ❖ Reached the R&D 100 Award Finals twice with the surrogate modeling solutions.

2021 – 2023

Postdoctoral Researcher, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

Developed the surrogate model ("LAROMance") for polycrystalline nuclear reactor materials by designing large-scale simulation pipelines and curating datasets from thousands of low length-scale simulations (utilizing state of the art HPC systems), establishing a framework for materials knowledge management and selection.

2019 – 2020

Design Engineer, ASML

DUV Def & Metrology group of Defectivity Department | Eindhoven, The Netherlands

Developed and validated advanced metrology systems for high-precision nanoscale defect detection on critical DUV lithography reticles, performing specialized cleanroom testing and data analysis under tight deadlines in collaboration with cross-functional and industrial teams.

2013 – 2019

PhD Researcher, Eindhoven University of Technology

Computational and Experimental Mechanics | Prof. Dr. Marc Geers & Dr. Johan Hoefnagels.

Philips Research, Materials Innovation Institute | Eindhoven, The Netherlands

Designed and implemented an integrated metrology framework combining finite element modeling, custom instrumentation, and microscale experimentation to quantify adhesion in microelectronic systems, resulting in five journal publications.

2017 – 2020

Account manager & Technical recruiter

Research-Square & EuFlex | Eindhoven, The Netherlands

Acquired and managed client accounts, and conducted technical recruitment of highly specialized engineering and PhD-level candidates for science and technology companies.

2018 – 2019

Lab technologist, Multi-Scale Laboratory

Eindhoven University of Technology | Eindhoven, The Netherlands

Computational and Experimental Mechanics

Managed an experimental mechanics laboratory, overseeing high-tech equipment including scanning electron microscopes and micro-mechanical test systems, while providing technical guidance to researchers and students and supporting experimental design in a high-demand environment.

2009 – 2013

Tutor of students

Eindhoven University of Technology, Mechanical Engineering Department | Freelance

Mentored bachelor student teams on engineering case studies, supporting balanced group dynamics, resolving team issues, and delivering constructive feedback.

Education

Eindhoven University of Technology, the Netherlands

Department of Mechanical Engineering

Division: Computational and Experimental Mechanics

Research Group: Mechanics of Materials

2013 – 2019

PhD.-research project

Title: "Identification of multi-layer interface systems: a full-field, micro-mechanical approach."

2013 – 2019

Graduate School of Engineering Mechanics

Advanced Mechanics Courses

2010 – 2013

MSc. Mechanical Engineering

Minor: Biomedical Engineering

2006 – 2010

BSc. Mechanical Engineering

2000 – 2006

Research-oriented secondary education (VWO)

St.-Janscollege, Hoensbroek, The Netherlands

Selected journal publications (peer-reviewed)

P. Robbe, A.P. Ruybalid, A. Hegde, C. Bonneville, H.N. Najm, L. Capolungo and C. Safta . A comparison of surrogate constitutive models for viscoplastic creep simulation of HT-9 steel. *Computational Materials Science*, 262, 114367 (2026). <https://doi.org/10.1016/j.commatsci.2025.114367>.

A.P. Ruybalid, A. Tallman, W. Wen, C. Matthews, L. Capolungo. Data-Driven Surrogate Modeling with Microstructure-Sensitivity of Viscoplastic Creep in Grade 91 Steel. *Integrating Materials and Manufacturing Innovation* 13, 895–914 (2024). <https://doi.org/10.1007/s40192-024-00377-z>

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers and J.P.M. Hoefnagels. Full-field identification of mixed-mode adhesion properties in a flexible, multi-layer microelectronic material system. *Engineering Fracture Mechanics*, 226, pp. 106879 (2020), <https://doi.org/10.1016/j.engfracmech.2020.106879>.

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers, J.P.M. Hoefnagels. An *In-Situ*, Micro-Mechanical Setup with Accurate, Tri-Axial, Piezoelectric Force Sensing and Positioning. *Exp Mech* 60, 713–725 (2020), <https://doi.org/10.1007/s11340-020-00583-8>.

A.P. Ruybalid, J.P.M. Hoefnagels, O. van der Sluis, M.P.F.H.L. van Maris and M.G.D. Geers. Mixed-mode cohesive zone parameters from integrated digital image correlation on micrographs only. *International Journal of Solids and Structures*, 156-157, pp. 179-193 (2019), <https://doi.org/10.1016/j.ijsolstr.2018.08.010>.